

5.2.2 Properties of the Definite Integral Worksheet

1. Suppose that f is integrable on the interval $[2, 9]$. Given that $\int_2^5 f(x) dx = 4$ and $\int_2^9 f(x) dx = 25$, evaluate the following definite integrals.

a) $\int_9^2 f(x) dx$

b) $\int_2^2 f(x) dx$

$$\int_9^2 f(x) dx = -\int_2^9 f(x) dx = -25$$

$$\int_2^2 f(x) dx = 0$$

c) $\int_2^5 [2f(x) + 3] dx$

d) $\int_5^9 f(x) dx$

$$\begin{aligned} \int_2^5 [2f(x) + 3] dx &= \int_2^5 2f(x) dx + \int_2^5 3 dx \\ &= 2 \int_2^5 f(x) dx + 3(5-2) = 2(4) + 3(3) = 17 \end{aligned}$$

$$\begin{aligned} \int_2^5 f(x) dx + \int_5^9 f(x) dx &= \int_2^9 f(x) dx \\ 4 + \int_5^9 f(x) dx &= 25 \\ \int_5^9 f(x) dx &= 21 \end{aligned}$$

2. Suppose that g is continuous at all real numbers and satisfies the following equations.

$$\int_{-2}^3 g(x) dx = 30$$

$$\int_{-2}^5 g(x) dx = 42$$

What is the value of $\int_5^3 [3g(x) - 2] dx$?

$$\begin{aligned} \int_5^3 [3g(x) - 2] dx &= \int_5^3 3g(x) dx - \int_5^3 2 dx = 3 \int_5^3 g(x) dx - 2(3-5) = -3 \int_3^5 g(x) dx - 2(-2) \\ &= -3 \left[\int_{-2}^5 g(x) dx - \int_{-2}^3 g(x) dx \right] + 4 = -3[42 - 30] + 4 = -36 + 4 = -32 \end{aligned}$$

3. Suppose that $w(x)$ is a continuous at all real numbers and satisfies the following equations.

$$\int_2^8 w(x) dx = 30 \qquad \int_6^{12} w(x) dx = -15 \qquad \int_8^{12} w(x) dx = 5$$

What is the value of $\int_2^6 [4w(x) + 25] dx$?

$$\int_2^6 [4w(x) + 25] dx = \int_2^6 4w(x) dx + \int_2^6 25 dx = 4 \int_2^6 w(x) dx + 25(6-2) = 4 \int_2^6 w(x) dx + 100$$

Note that:

$$\begin{aligned} \int_2^6 w(x) dx + \int_6^{12} w(x) dx &= \int_2^8 w(x) dx + \int_8^{12} w(x) dx \\ \Rightarrow \int_2^6 w(x) dx - 15 &= 30 + 5 \\ \Rightarrow \int_2^6 w(x) dx &= 50 \end{aligned}$$

Therefore:

$$\int_2^6 [4w(x) + 25] dx = 4 \int_2^6 w(x) dx + 100 = 4(50) + 100 = 300$$

4. Use the properties of integrals to estimate the integral $\int_{-1}^1 \sqrt{1+x^2} dx$.

We notice that if $-1 \leq x \leq 1$ then we can say that:

$$-1 \leq x \leq 1 \Rightarrow 0 \leq x^2 \leq 1 \Rightarrow 1 \leq 1+x^2 \leq 2 \Rightarrow \sqrt{1} \leq \sqrt{1+x^2} \leq \sqrt{2} \Rightarrow 1 \leq \sqrt{1+x^2} \leq \sqrt{2}$$

Therefore, by *Property 10* of integrals we have:

$$\begin{aligned} \int_{-1}^1 1 dx &\leq \int_{-1}^1 \sqrt{1+x^2} dx \leq \int_{-1}^1 \sqrt{2} dx \\ 1(1-(-1)) &\leq \int_{-1}^1 \sqrt{1+x^2} dx \leq \sqrt{2}(1-(-1)) \\ 1(2) &\leq \int_{-1}^1 \sqrt{1+x^2} dx \leq \sqrt{2}(2) \\ 2 &\leq \int_{-1}^1 \sqrt{1+x^2} dx \leq 2\sqrt{2} \end{aligned}$$

5. Use properties of integrals to demonstrate how one could determine that the following inequalities hold.

$$0.20 \leq \int_9^{13} \frac{1}{4\sin^4(x^{12})+16} dx \leq 0.25$$

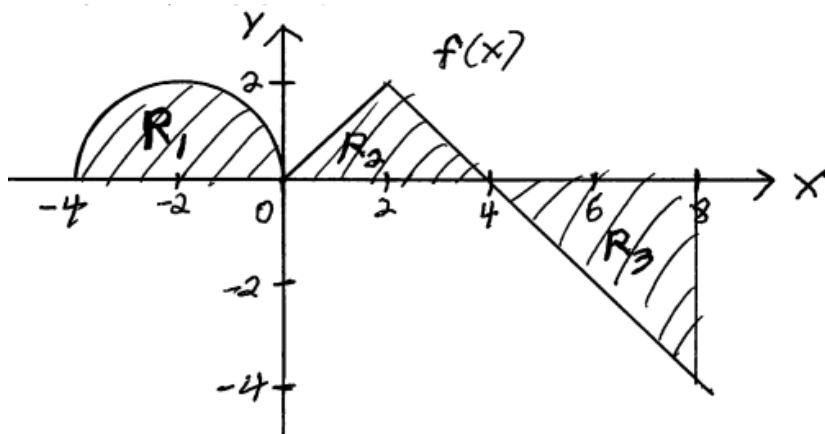
We notice that if $9 \leq x \leq 13$ then we can say that:

$$\begin{aligned} 9 \leq x \leq 13 &\Rightarrow -1 \leq \sin(x^{12}) \leq 1 \Rightarrow 0 \leq \sin^4(x^{12}) \leq 1 \Rightarrow 0 \leq 4\sin^4(x^{12}) \leq 4 \\ &\Rightarrow 16 \leq 4\sin^4(x^{12})+16 \leq 20 \Rightarrow \frac{1}{20} \leq \frac{1}{4\sin^4(x^{12})+16} \leq \frac{1}{16} \end{aligned}$$

Therefore, by *Property 10* of integrals we have:

$$\begin{aligned} \int_9^{13} \frac{1}{20} dx &\leq \int_9^{13} \frac{1}{4\sin^4(x^{12})+16} dx \leq \int_9^{13} \frac{1}{16} dx \\ \frac{1}{20}(13-9) &\leq \int_9^{13} \frac{1}{4\sin^4(x^{12})+16} dx \leq \frac{1}{16}(13-9) \\ \frac{1}{20}(4) &\leq \int_9^{13} \frac{1}{4\sin^4(x^{12})+16} dx \leq \frac{1}{16}(4) \\ 0.20 &\leq \int_9^{13} \frac{1}{4\sin^4(x^{12})+16} dx \leq 0.25 \end{aligned}$$

6. The graph of $f(x)$ is shown below. Evaluate $\int_{-4}^8 f(x) dx$.



$$\int_{-4}^8 f(x) dx = \text{area}(R_1) + \text{area}(R_2) + \text{area}(R_3)$$

$$= \frac{1}{2}\pi(2)^2 + \frac{1}{2}(4)(2) + \frac{1}{2}(4)(-4) = 2\pi + 4 - 8 = 2\pi - 4$$

7. Rounded off to one decimal place the integral $\int_{10}^{15} \left(\frac{2^{2-\sin^2(x)}}{5} \right) dx$ is equal to one of the choices below. Use

properties of integrals to circle the correct choice. Show enough work to justify your answer. Do not attempt to evaluate the integral directly.

- | | | | | |
|---------|---------|---------|---------|----------|
| (a) 0.3 | (b) 0.8 | (c) 1.1 | (d) 1.9 | (e) 2.7 |
| (f) 4.3 | (g) 6.8 | (h) 7.2 | (i) 9.7 | (j) 14.9 |

We notice that if $9 \leq x \leq 13$ then we can say that:

$$\begin{aligned}
 10 \leq x \leq 15 &\Rightarrow 0 \leq \sin^2(x) \leq 1 \Rightarrow 1 \leq 2 - \sin^2(x) \leq 2 \Rightarrow 2^1 \leq 2^{2-\sin^2(x)} \leq 2^2 \\
 &\Rightarrow \frac{2}{5} \leq \frac{2^{2-\sin^2(x)}}{5} \leq \frac{4}{5}
 \end{aligned}$$

Therefore, by *Property 10* of integrals we have:

$$\begin{aligned}
 \int_{10}^{15} \frac{2}{5} dx &\leq \int_{10}^{15} \left(\frac{2^{2-\sin^2(x)}}{5} \right) dx \leq \int_{10}^{15} \frac{4}{5} dx \\
 \frac{2}{5}(15-10) &\leq \int_{10}^{15} \left(\frac{2^{2-\sin^2(x)}}{5} \right) dx \leq \frac{4}{5}(15-10) \\
 \frac{2}{5}(5) &\leq \int_{10}^{15} \left(\frac{2^{2-\sin^2(x)}}{5} \right) dx \leq \frac{4}{5}(5) \\
 2 &\leq \int_{10}^{15} \left(\frac{2^{2-\sin^2(x)}}{5} \right) dx \leq 4
 \end{aligned}$$

The only possible choice that falls within 2 and 4 is (e).